

Annotating TikTok Videos with LLMs

How I plan to use Apollo

Francesco Bailo

The project

Does casual attention to mild content push recommenders toward extreme content?

- Testing the “manosphere pipeline” hypothesis on TikTok
- Data: ~407K timestamped video exposures from 142 participants (browser-tracing)
- Realistic download budget: **~10,000–20,000 videos** — a subset of participants, each with their full, contiguous viewing history
- Each video needs a **topic-severity label** (3-tier taxonomy, Fisher et al. 2026) to build exposure sequences for a transition model
- Ground truth: **2,414 videos already hand-coded** (Fisher et al.) — used to validate LLM labels
- **Current bottleneck is TikTok downloads, not classification** — GPU annotation is fast once videos are in hand

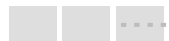
Why I need GPUs

- Manual coding doesn't scale even to 10,000–20,000 videos — LLM annotation is what makes this feasible at all
- **Can't subsample within a participant's history:** the transition model needs each included participant's *entire, contiguous* viewing sequence
- So we select **fewer participants**, not fewer videos per participant
- Approach: **open-weight LLMs via Ollama**, zero/few-shot classification into the 3-tier taxonomy (cultural touchpoints → masculine status → degrading health)

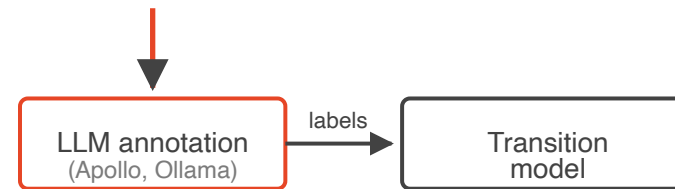
142 participants

A  dropped

B  **full sequence kept**

C  dropped

: (139 more participants)



Each square = one video exposure, in timestamp order.
Sequences are never cut mid-participant — only whole participants are included or excluded.

Selecting participants, not subsampling within a timeline

Early local test: Gemma 4

Ran 6 sample TikTok videos end-to-end (frame extraction → vision-language model → structured JSON) with `gemma4:e2b`, locally on an M1 Mac — fast enough, and annotation quality looks promising.

Placeholder prompt for pipeline testing — not the final codebook.

```
{  
  "description": "<1-2 sentence factual description of what happens>",  
  "category": "<one of: dance, comedy, food, beauty, fashion, education,  
             sports, music, pets, gaming, lifestyle, news_politics, other>",  
  "on_screen_text": "<any visible captions/text overlays, or empty string>",  
  "notable_objects": ["<short list of key objects/people/settings>"],  
  "sentiment": "<one of: positive, neutral, negative>"  
}
```

Sample output



```
{  
  "description": "A man performs various poses, including a  
                 handstand on a concrete wall and a stretch  
                 on the sidewalk, in an urban outdoor setting.",  
  "category": "lifestyle",  
  "on_screen_text": "",  
  "notable_objects": ["man", "concrete wall", "sidewalk",  
                     "street signs", "urban background"],  
  "sentiment": "neutral"  
}
```